



Department of Computer Science and Engineering

Academic Year 2021 – 2022(Odd / ~~Even~~ Semester)

Degree, Semester & Branch: B.E, VII & CSE

Course Code & Title: CS8791 & Cloud Computing

Name of the Faculty member (s): Dr.M.Gomathy Nayagam

Innovative Practice Description

- **Unit / Topic:** Unit V /Map Reduce

- **Course Outcome:** CO 5

- **Topic Learning Outcome:** TLO13

- **Activity Chosen:** Flipped Class

- **Justification:**

Apache Hadoop is one of the big data analytics tools which will be used by the most of the industry. It is highly reliable, scalable, distributed processing tool for big data. HDFS is one of component in Apache hadoop. The primary objective of the flipped class is to allow the students for self-paced learning, dive deeper into the concept, enhance the communiqué skill, inventiveness and analytic skill. Hence this activity will help the students to explore knowledge in Mapreduce concepts well and good.

- **Time Allotted for the Activity:** 40 Minutes

- **Details of the Implementation:**

- Video lecture corresponding to the particular topic was hosted in online one week before.
- Students were instructed to watch the video and took the notes according to their understanding level in their home and come with preparation for discussion on one particular day in the class room.
- The students were segmented as a group in the class during the day of discussion.
- We gave 10 minutes for discussion among the group member regarding the video lecture they watched earlier.
- Each group collected all the important points related to the topic from those group members and to be allotted 15 minutes time for on board presentation and discussion about the topic in front of the whole class with their corresponding group members.
- Corresponding subject faculty monitored each group's presentation.
- Course Instructor selected one among the student in the class and asked the particular student to summarize all groups' presentation for next 5 minutes

• **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO 3	3	2	2	2	2	-	-	1	1	2	-	1	2	3	1

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO 1	PO 8	PO 9	PO 10	PSO2
	3	1	2	1	3
Justification for correlation	Students could be able to apply basic theory and principles of Map reduce programming model in Hadoop	Based on the theoretical knowledge gathered, group of students prepared their own points without copying other groups points.	Students could be able to recollect the various map reduce programming model thorough their group	Students could able to get the knowledge clearly about the map reduce	Students could be able to write simple word count problem using map reduce programming model in hadoop.

• **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students are enjoyed to prepare the concept through self-learning. They felt that this activity is very helpful to them for improving their self-learning ability.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- The students can able to explore the knowledge in map reduce programming model concepts by improving their self-paced learning and get more key points from their peer groups when they are doing collaborative discussion among them.

- ❖ ***Challenges faced in implementation:***

- Difficult to make the individual to watch the video which is already hosted in canvas.
- Difficult to Select the video for the activity and create the group which contains heterogeneous students.
- Space for conducting this activity inside the class room is not so good.

References:

- ❖ <https://www.ritrjpm.ac.in/images/computer-science/CS8603-flippedClassroom.pdf>
- ❖ https://www.ritrjpm.ac.in/images/computer-science/32_CS6703_Flippedclassroom.pdf
- ❖ Flipped Learning Network (FLN). (2014) The Four Pillars of F-L-I-P™
- ❖ Stephanie butler velegol, sarah e. Zappe, emily mahoney, The Evolution of a Flipped Classroom: Evidence-Based Recommendations, winter 2015, pp.1-37
- ❖ Dian Mita Nurhayati and Hartono, Implementation of Cooperative Learning Model Type with RME Approach to Understanding of Mathematical Concept Student State Junior High School in Pekanbaru, AIP Conference Proceedings 1848, 040002, 2017, <https://doi.org/10.1063/1.4983940>
- ❖ Micheal M van Wyk, The Effects of the Cooperative Learning Method on Student Achievement, Attitude and Motivation in Economics Education, Journal of Social Science pp. 261-270, 2017, <https://doi.org/10.1080/09718923.2012.11893104>

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